

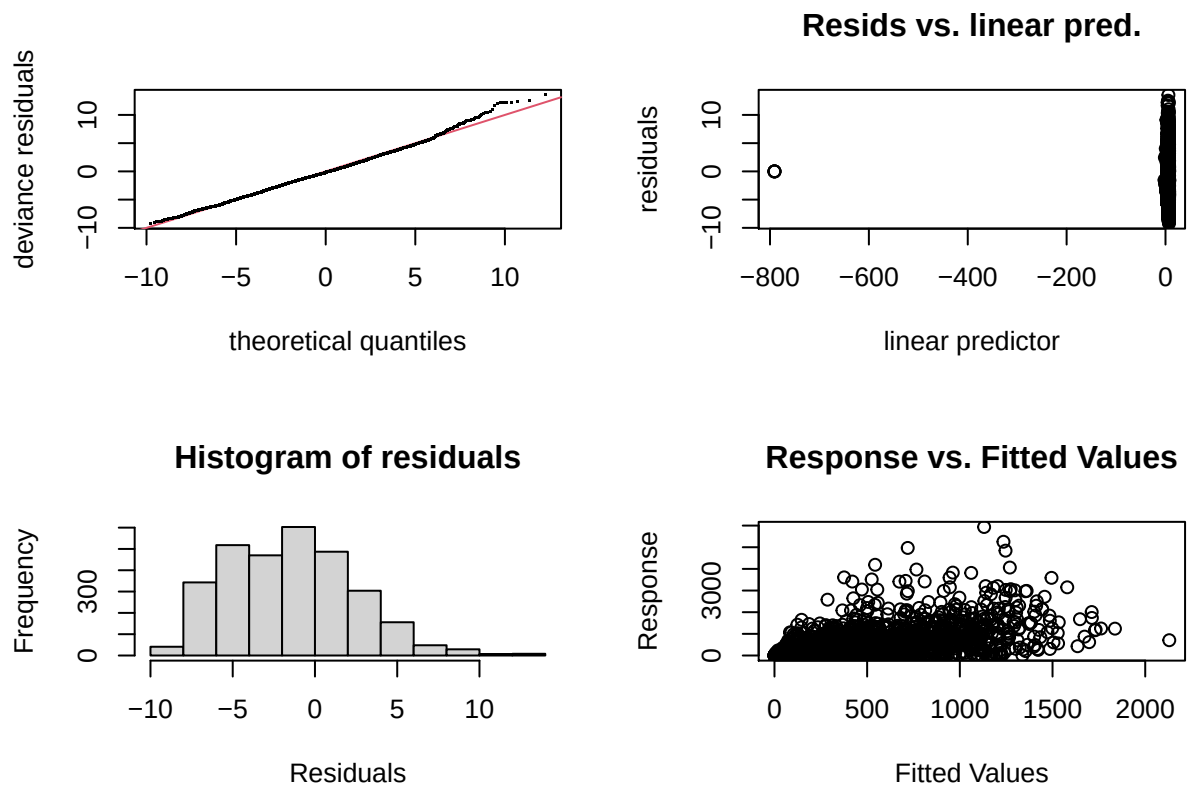
GAM diagnostics

GAM

The best model from the preliminary GAMs included most covariates:

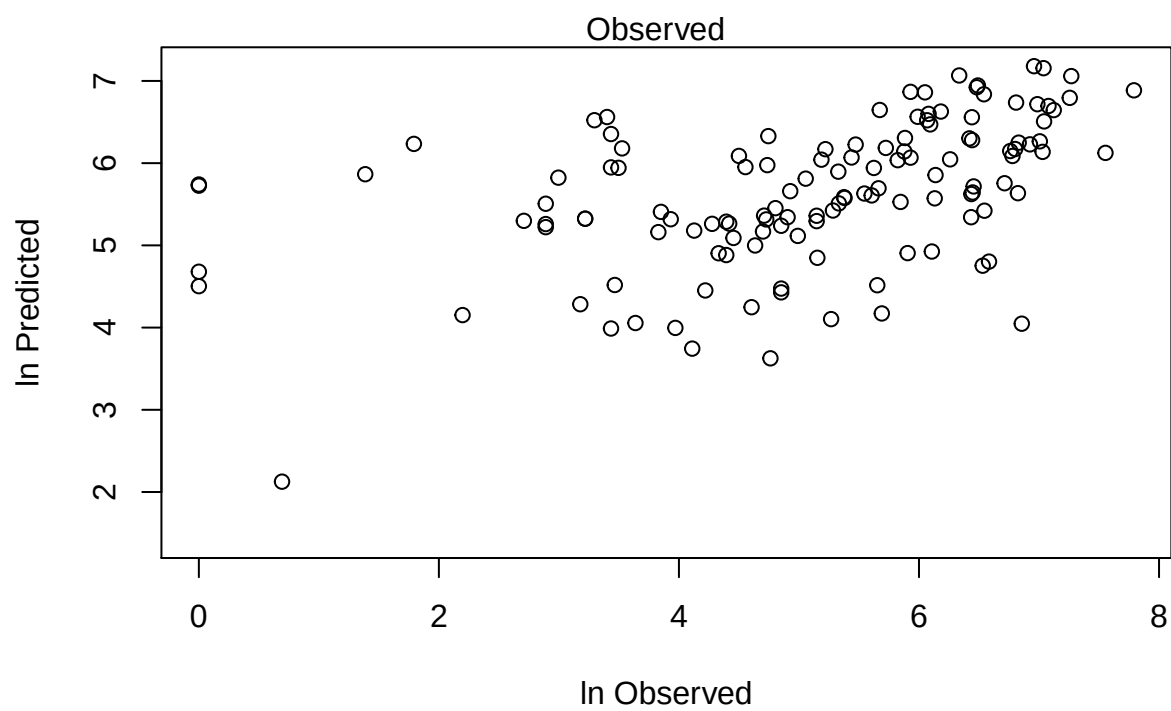
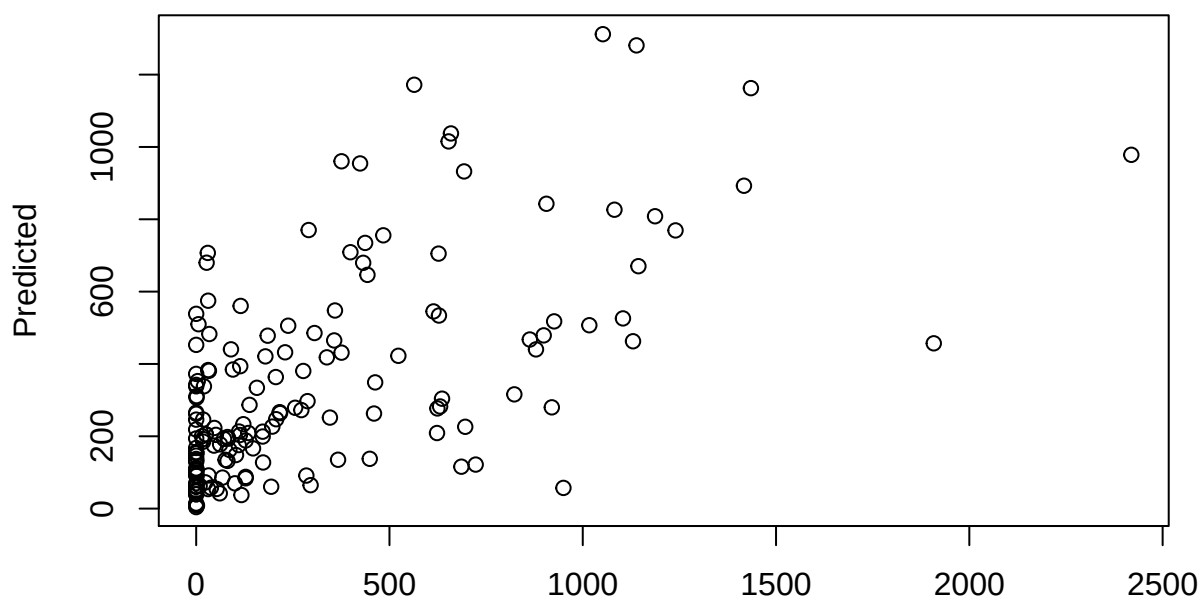
```
gam(Juv.km ~ -1 + as.factor(STRM_ORDER) + s(CLASS_Rank,k=3) +  
      s(StrmSlope,k=4) +  
      s(MaxGradD,k=4) +  
      s(WidthM,k=4) +  
      s(OUT_DIST,k=4) +  
      s(Barriers,k=4) +  
      s(SolMean,k=4) +  
      s(MWMT_Index,k=4) +  
      s(W3Dppt,k=4) +  
      s(IP_COHO,k=4) +  
      s(PopGrp,bs="re"),  
  data=train, family = "tw")
```

Residual diagnostics from best model

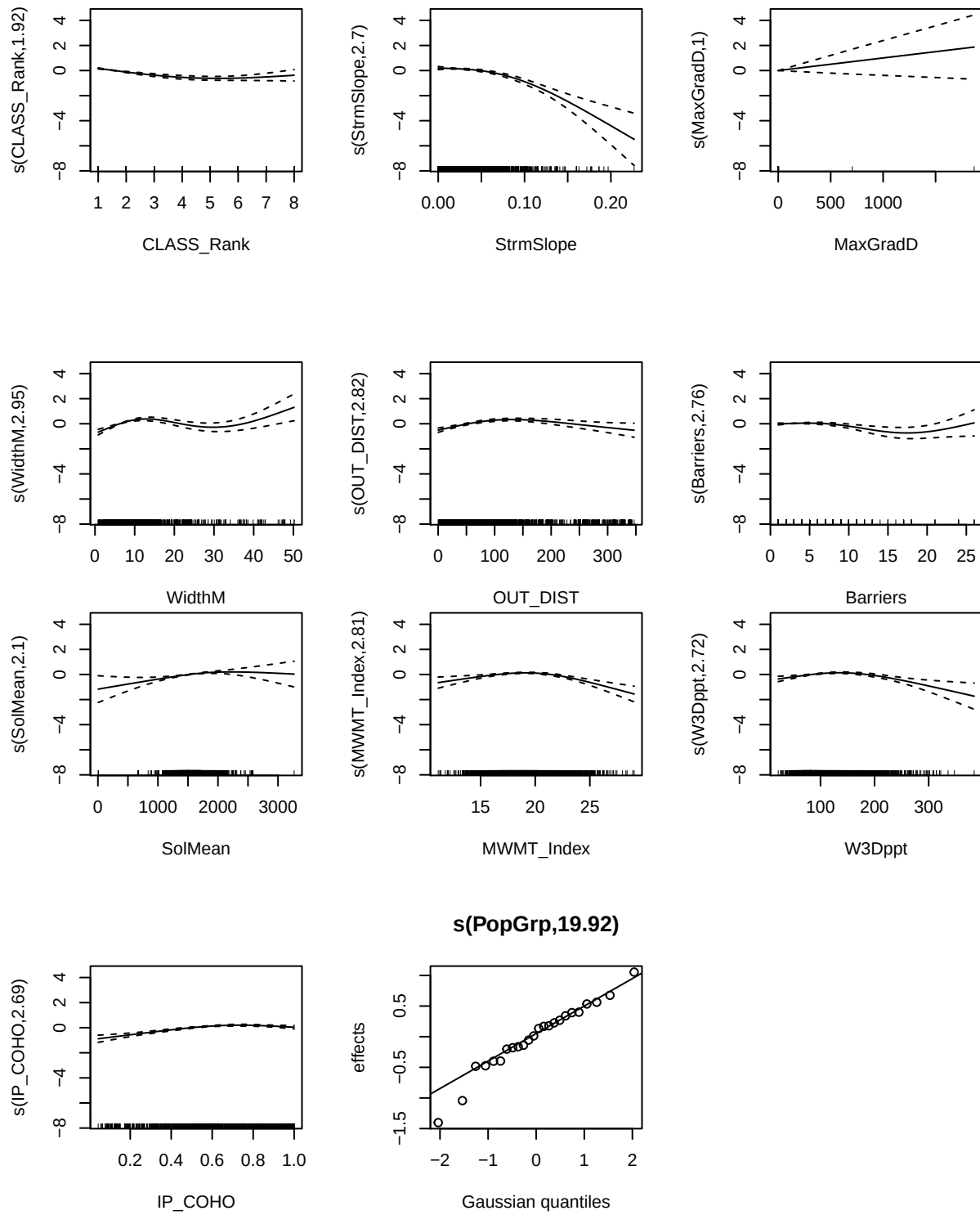


```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-0.00492374,0.003419895]
## (score 18108.13 & scale 13.7169).
## Hessian positive definite, eigenvalue range [0.002044972,4820.773].
## Model rank = 71 / 71
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##          k'   edf k-index p-value
## s(CLASS_Rank) 2.00 1.92  0.60 <2e-16 ***
## s(StrmSlope)  3.00 2.70  0.62 <2e-16 ***
## s(MaxGradD)   3.00 1.00  0.60 <2e-16 ***
## s(WidthM)     3.00 2.95  0.65 <2e-16 ***
## s(OUT_DIST)   3.00 2.82  0.62 <2e-16 ***
## s(Barriers)   3.00 2.76  0.62 <2e-16 ***
## s(SolMean)    3.00 2.10  0.61 <2e-16 ***
## s(MWMT_Index) 3.00 2.81  0.84  0.035 *
## s(W3Dppt)     3.00 2.72  0.79 <2e-16 ***
## s(IP_COHO)    3.00 2.69  0.66 <2e-16 ***
## s(PopGrp)     24.00 19.92 NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

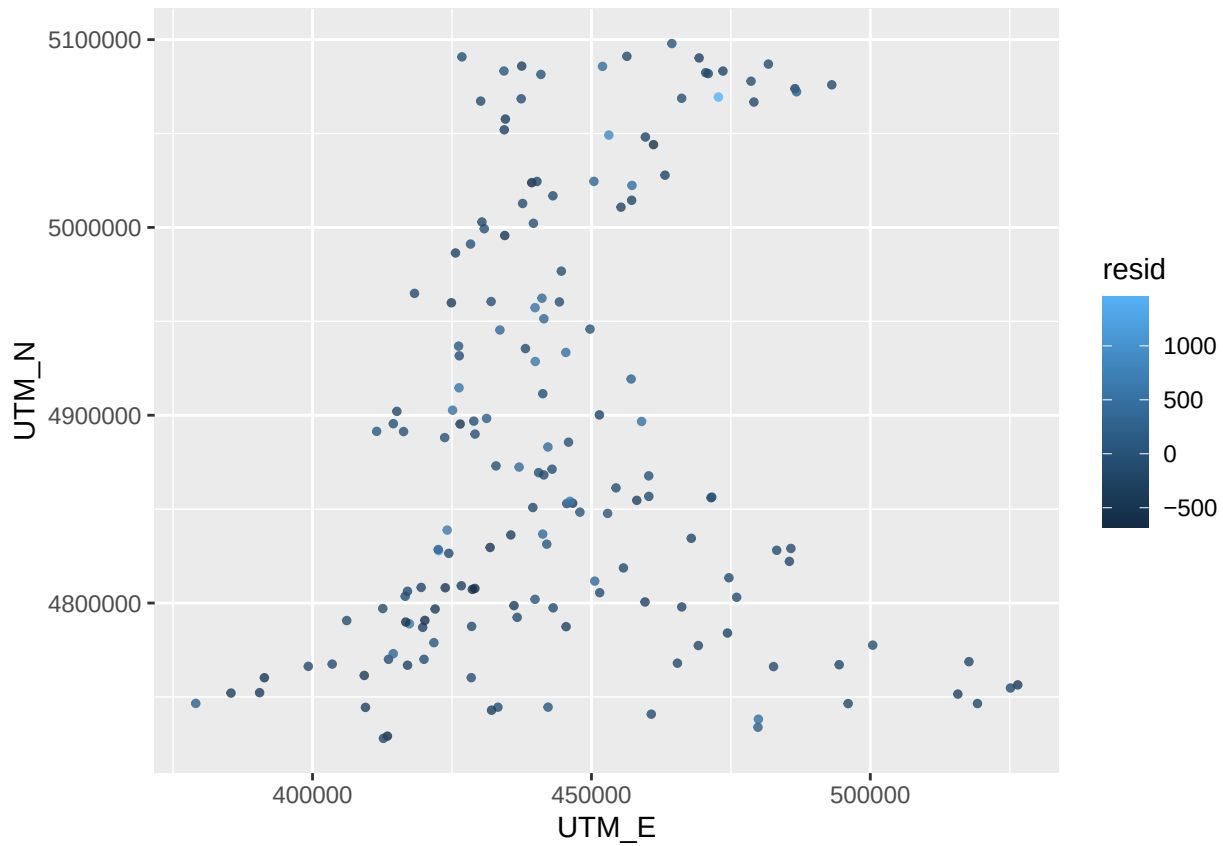
Out of sample prediction



Smooths



Residual spatial autocorrelation?



```
### Moran's I
## $observed
## [1] 0.08356264
##
## $expected
## [1] -0.00617284
##
## $sd
## [1] 0.01744109
##
## $p.value
## [1] 2.674314e-07
```